CS 532, Fall 2017: Quiz 9

NAME:

Problem 1. How can you convert a pair of depthmaps into a 3D point cloud expressed in common reference frame? What information would you need? Explain your answer and make an illustrative sketch.

Problem 2. Consider a tetrahedron with 3D vertices at the following coordinates:

$$v_1 = (1, 1, 0)\ v_2 = (1, 2, 0)\ v_3 = (1, 1, 1)\ v_4 = (2, 1, 0)$$

a) List all the surface normal vectors (i.e one for each of the faces) and make a sketch.

b) Consider a rotation of 90 degrees around the Z axis has been applied to all the vertices. List all the modified normal vectors.